



Lexical exceptionality in paradigm-specific learning: Modeling stem-final obstruent alternations in Korean verbs and adjectives

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- **Lexical Exceptionality**

- Learners must identify exceptional forms and posit item-specific grammars departing from a general pattern.

- **Within Optimality Theory**

- **Cophonology** (Inkelas and Zoll, 2003)
 - Separate rankings for different morphological classes
Class X: Constraint A $\gg$ Constraint B
Class Y: Constraint B $\gg$ Constraint A
- **Lexical Indexation** (Pater, 2000)
 - Constraints indexed to lexical classes
Constraint A_{class X} $\gg$ Constraint B $\gg$ Constraint A_{class Y}

• Too Many Solutions Problem

- Exuberance in solving markedness violations (McCarthy, 2002)
- Different solutions to *NC̥ (Pater, 1999)
 - /mp/ → [mb]: violates IDENT(VOICE)
 - /mp/ → [bp]: violates IDENT(NASAL)
 - /mp/ → [m]/[p]: violates MAX-C

• Lexical Exceptionality X Multiple Repair Strategies

- When multiple repair strategies coexist for lexically exceptional patterns, how do learners simultaneously identify exceptional items and learn the corresponding repair strategy?

Korean Stem-final Conjugation

- A useful test case for the interaction of lexical exceptionality and multiple repair strategies
- Three alternating classes undergo intervocalic lenition, each exhibiting a distinct repair strategy:
 - /p/ → [w]
 - /t/ → [ɾ]
 - /s/ → ∅
- Non-alternating classes remain faithful.

Class	Stem	Conjugation
/p/-alternating	/ka.kap/ 'close'	[ka.kɑ.w-Λ]
/p/-non-alternating	/ip/ 'wear'	[i.p-Λ]
/t/-alternating	/kjɛ.dat/ 'realize'	[kjɛ.da.ɾ-a.jo]
/t/-non-alternating	/mut/ 'bury'	[mu.t-Λ]
/s/-alternating	/nas/ 'better'	[na-a.do]
/s/-non-alternating	/us/ 'laugh'	[u.s-Λ]

Korean Stem-final Conjugation

- Suffixes provide the phonological environment triggering alternation.
 - Before consonant-initial suffixes, the intervocalic environment is absent.
 - The contrast between alternating and non-alternating classes is no longer observable.
- *Lexical idiosyncrasy resides in the stem*

Class	Stem	Conjugation
/p/-alternating	/ka.kap/ 'close'	[ka.kap-ta]
/p/-non-alternating	/ip/ 'wear'	[ip-ta]
/t/-alternating	/kje.dat/ 'realize'	[kje.dat-ta]
/t/-non-alternating	/mut/ 'bury'	[mut-ta]
/s/-alternating	/nas/ 'better'	[nas-ta]
/s/-non-alternating	/us/ 'laugh'	[us-ta]

- Lee (2013) analyzes Korean stem-final obstruent alternations using lexical indexation.
- A shared markedness constraint:

$$*V-OBS-V$$

- Ranking paradox:

$$*V-OBS-V \gg \text{FAITH} \quad (\text{alternating})$$

$$\text{FAITH} \gg *V-OBS-V \quad (\text{non-alternating})$$

- Solution: constraint cloning (Pater, 2009)

$$*V-OBS-V_{\text{Alter}} \gg \text{FAITH} \gg *V-OBS-V_{\text{Non-Alter}}$$

- **Unaddressed:** How do paradigm-specific instantiations of the lexical-indexation schema combine into a single grammar?
- Different repair strategies violate different FAITH constraints:
 - /p/-alternating stems violate ID-C(ONSONANTAL)
 - /t/-alternating stems violate ID-LAT(ERAL)
 - /s/-alternating stems violate MAX(-C)
- **Expanded Ranking:** FAITH constraint implicated in a given repair must be outranked by those associated with the competing repair strategies.

$$*V-OBS-V_P^{Alt} \gg ID-LAT, MAX \gg ID-C \gg *V-OBS-V_P^{NonAlt}$$
$$*V-OBS-V_T^{Alt} \gg ID-C, MAX \gg ID-LAT \gg *V-OBS-V_T^{NonAlt}$$
$$*V-OBS-V_S^{Alt} \gg ID-C, ID-LAT \gg MAX \gg *V-OBS-V_S^{NonAlt}$$

Lexically-Scaled MaxEnt

A probabilistic implementation of constraint indexation (Linzen et al., 2013; Hughto et al., 2019)

$$\mathcal{H}_i = \sum_{\gamma \in \Gamma} \left(w_\gamma + \sum_{m \in \mu_i} s_{\gamma m} \right) v_{\gamma i}$$

(Γ : constraint set; $\mathcal{H}_i$: candidate harmony; w_γ : general constraint weight; $s_{\gamma m}$: morpheme-specific scaling factor; $v_{\gamma i}$: violations of γ by candidate i)

The additive penalty $s_{\gamma m}$ serves as a probabilistic analogue of lexical indexation.

1. Alternating classes receive increased scaling on *V-OBS-V, yielding *V-OBS-V $\gg$ FAITH. Non-alternating classes receive no comparable scaling, consistent with FAITH $\gg$ *V-OBS-V.

2. Repair-specific FAITH constraints receive the smallest scaling:
 - ID-C(ONSONANTAL) (/p/-alternation)
 - ID-LAT(ERAL) (/t/-alternation)
 - MAX(-C) (/s/-alternation)
3. Only idiosyncratic stems receive nonzero scaling; suffixes remain unscaled.

	*V-OBS-V	MAX	ID-LAT	ID-C
General weights	4	6	6	6
p-alternating	6	3	2	0
p-non-alternating	0	1	1	1
t-alternating	6	3	0	4
t-non-alternating	0	1	1	1
s-alternating	6	0	3	3
s-non-alternating	0	1	1	1
suffixes	0	0	0	0

Idealized learning outcome

- **Toy stems:** generated in a CV-/p,t,s/ structure
 - Reflect Korean phonotactics (no consonant clusters)
 - Segment inventories based on the NIKL (National Institute of Korean Language) Romanization system
 - $C \in \{g, kk, k, d, tt, t, b, pp, p, j, jj, ch, s, ss, h, n, m, r\}$
 - $V \in \{a, eo, o, u, eu, i, ae, e, oe, wi\}$
 - Stem class proportions mirror type frequencies reported in Lee (2013)

p-alternating	68 (72.3%)
p-non-alternating	5 (5.3%)
t-alternating	5 (5.3%)
t-non-alternating	7 (7.5%)
s-alternating	5 (5.3%)
s-non-alternating	4 (4.3%)

Stem Class Distribution in the Sejong Corpus

Complete inventory of toy stems

Class	Stems
p-alternating	neop- ppaep- sep- ssep- gop- bwip- seup- dep- dop- baep- nep- kkeup- moep- kkip- kkop- ssaep- chap- rep- kop- teup- cheop- tep- heup- rwip- jjwip- paep- bop- bup- mip- dwip- choep- sop- swip- pwip- jjeop- keop- soep- ssap- taep- mwip- rup- chwip- bep- peup- ppep- chop- keup- geup- nip- ppip- saep- noep- kkap- jjup- gwip- jjep- meup- ttop- jep- daep- joep- mop- mep- ssoep- ttaep- reop- jjop- ppap-
p-non-alternating	jaep- kkep- jjip- rap- toep-
t-alternating	tat- jat- ket- pot- chit-
t-non-alternating	ppeut- goet- jot- pput- nat- kkut- sit-
s-alternating	ges- kkaes- hoes- kies- nos-
s-non-alternating	poes- meos- jjaes- ches-

Experiment

- **Toy suffixes:** designed to control exposure to intervocalic lenition environments
 - Alternation-triggering (-V) and non-triggering (-CV) suffixes were equally represented
 - Initial inventory consisted of two suffixes: *-a* and *-ta*
 - Additional V-/CV pairs were incrementally introduced
 - Vowels were drawn from the same monophthong inventory used in stem generation

Condition	Suffix inventory
2-suffix	<i>-ta</i> , <i>-a</i>
4-suffix	<i>-ta</i> , <i>-a</i> , <i>-te</i> , <i>-e</i>
6-suffix	<i>-ta</i> , <i>-a</i> , <i>-te</i> , <i>-e</i> , <i>-tu</i> , <i>-u</i>
8-suffix	<i>-ta</i> , <i>-a</i> , <i>-te</i> , <i>-e</i> , <i>-tu</i> , <i>-u</i> , <i>-to</i> , <i>-o</i>
10-suffix	<i>-ta</i> , <i>-a</i> , <i>-te</i> , <i>-e</i> , <i>-tu</i> , <i>-u</i> , <i>-to</i> , <i>-o</i> , <i>-ti</i> , <i>-i</i>

Balanced Condition

- The /p/-paradigm was excluded to create a balanced distribution of repair frequencies.
- This removes the dominant glide-formation pattern associated with /p/-alternation.
- Two repair strategies are available for resolving *V-OBS-V:
 1. Lateralization (violating ID-LAT)
 2. Deletion (violating MAX)
- Each UR is paired with three candidate SRs:

UR	SR	%	*V-OBS-V	MAX	ID-LAT
/tat-a _T -Alter/	(a) [tata]	0.0	1	0	0
	(b) [tara]	1.0	0	0	1
	(c) [taa]	0.0	0	1	0
/ges-ta _S -Alter/	(a) [gesta]	1.0	0	0	0
	(b) [gerta]	0.0	0	0	1
	(c) [geta]	0.0	0	1	0

Example simulation data in the balanced condition

Imbalanced Condition

- The /p/-paradigm is included, yielding a skewed distribution of repair frequencies.
- This introduces the dominant glide-formation pattern associated with /p/-alternation.
- In addition to lateralization and deletion, glide formation (violating ID-C) becomes available.
- Each UR is paired with four candidate SRs:

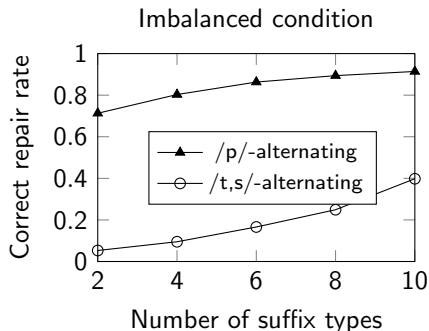
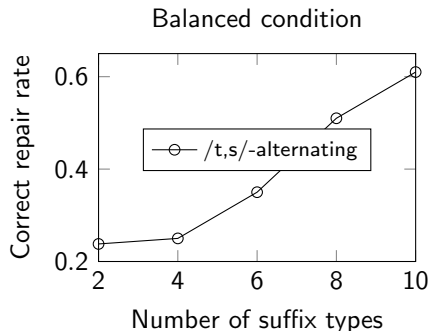
UR	SR	%	*V-OBS-V	MAX	ID-C	ID-LAT
/neop-a _P -Alter/	(a) [neopa]	0.0	1	0	0	0
	(b) [neowa]	1.0	0	0	1	0
	(c) [neola]	0.0	0	0	0	1
	(d) [neoa]	0.0	0	1	0	0
/tat-a _T -Alter/	(a) [tata]	0.0	1	0	0	0
	(b) [taja]	0.0	0	0	1	0
	(c) [tara]	1.0	0	0	0	1
	(d) [taa]	0.0	0	1	0	0

Example simulation data in the imbalanced condition

Results

Generalization improves with increasing suffix type frequency.

Evaluation focuses on repair accuracy of alternating stems in intervocalic contexts, i.e., when followed by -V suffixes.



→ This progression is reflected in the distribution of scaling weights.

Results

Stage 1: When given only few suffix types, scaling weights are misattributed to non-idiosyncratic suffixes.

	*V-OBS-V	MAX	ID-LAT
General weights	0	0.79	0.79
t-alternating stems	0	0	0
t-non-alternating stems	0	0	0
s-alternating stems	0	0	0
s-non-alternating stems	0	0	0
-V suffix	0	0	0
-CV suffix	0	2.16	2.16

Balanced condition (2 suffixes)

	*V-OBS-V	MAX	ID-LAT	ID-C
General weights	0.69	2.6	2.6	0
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	0
t-alternating stems	0	0	0	0
t-non-alternating stems	0	0	0	0
s-alternating stems	0	0	0	0
s-non-alternating stems	0	0	0	0
-V suffix	0.69	0	0	0
-CV suffix	0	1.92	1.92	4.51

Imbalanced condition (2 suffixes)

- In the 2-suffix condition, each stem is observed only twice, whereas the same suffixes occur across all stem classes.
- Scaling weights are consequently assigned to suffixes rather than stems.

Results

Stage 2: Scaling weights shift from suffixes to stems, enabling the learner to distinguish alternating and non-alternating classes.

	*V-OBS-V	MAX	ID-LAT
General weights	2.2	3.26	3.26
t-alternating stems	0.37	0	0
t-non-alternating stems	0	0	0
s-alternating stems	0.37	0	0
s-non-alternating stems	0	0	0
-V suffix	0	0	0
-CV suffix	0	0	0

Balanced condition (4 suffixes)

	*V-OBS-V	MAX	ID-LAT	ID-C
General Weights	4.98	6.29	6.29	2.71
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	2.54
t-alternating stems	0.61	0	0	2.88
t-non-alternating stems	0	0	0	2.54
s-alternating stems	0.61	0	0	2.88
s-non-alternating stems	0	0	0	2.54
-V suffix	0	0	0	0
-CV suffix	0	0	0	1.53

Imbalanced condition (6 suffixes)

- As suffix diversity increases, stems occur in a wider range of contexts, making them more likely to be lexicalized.
- The learner first distinguishes alternating from non-alternating classes.
- Exception: /p/-alternating stems are not lexicalized.

Stage 3: Paradigm-specific patterns are distinguished among alternating classes

	*V-OBS-V	MAX	ID-LAT
General weights	1.72	4.04	4.04
t-alternating stems	3.43	1.14	0
t-non-alternating stems	0	0	0
s-alternating stems	3.43	0	1.14
s-non-alternating stems	0	0	0
-V suffix	0	0	0
-CV suffix	0	0	0

Balanced condition (10 suffixes)

	*V-OBS-V	MAX	ID-LAT	ID-C
General Weights	7.11	8.98	8.98	4.46
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	3.76
t-alternating stems	2.55	0.69	0	5.2
t-non-alternating stems	0	0	0	3.76
s-alternating stems	2.55	0	0.69	5.2
s-non-alternating stems	0	0	0	3.76
-V suffix	0	0	0	0
-CV suffix	0	0	0	0

Imbalanced condition (10 suffixes)

- Alternating classes are further differentiated according to their repair strategy.
- /t/- and /s/-alternating classes establish distinct rankings between ID-LAT and MAX.
- Exception: /p/-alternating stems remain unlexicalized.

Exception: Statistical dominance prevents lexicalization of /p/-alternating stems.

Stem class	Type freq.
p-alternating	68 (72.3%)
p-non-alternating	5 (5.3%)
t-alternating	5 (5.3%)
t-non-alternating	7 (7.5%)
s-alternating	5 (5.3%)
s-non-alternating	4 (4.3%)

	*V-OBS-V	MAX	ID-LAT	ID-C
General weights	7.11	8.98	8.98	4.46
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	3.76
t-alternating stems	2.55	0.69	0	5.2
t-non-alternating stems	0	0	0	3.76
s-alternating stems	2.55	0	0.69	5.2
s-non-alternating stems	0	0	0	3.76
-V suffix	0	0	0	0
-CV suffix	0	0	0	0

- The /p/-alternating class overwhelmingly dominates the stem distribution (72.3%).
- /p/ → [w] alternation is treated as a default repair strategy rather than a lexically idiosyncratic pattern.
 - The /p/-alternating stems remain unscaled.
 - ID-C receives high scaling weights in the remaining classes.

Sensitivity to lexical statistics

- The model adopts a *least-cost* strategy in tracking lexical exceptionality.
 - Statistically dominant morpheme classes are learned as the general pattern.
 - Exceptionality is not always localized to the true exceptional class:
 - /p/-alternating stems are not treated as idiosyncratic.
 - Under limited morphological evidence, scaling weights are assigned to suffixes.
- How are suffix frequencies distributed in natural language?

Absolute suffix type frequency seems unlikely to be the bottleneck

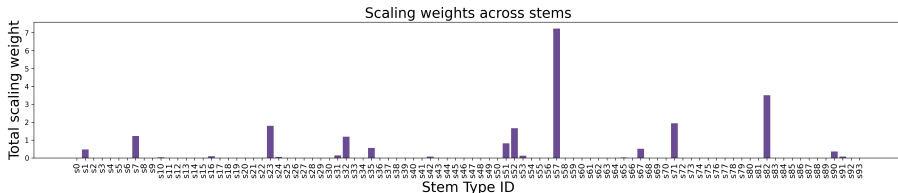
- The Korean Google Treebank contains 29 C-initial and 33 V-initial suffix types, far exceeding the largest suffix inventories considered in the simulation.

Context	Suffix inventory
-CV	-geona, -ge, -get, -go, -goyo, -guyo, -gi, -gillae, -na, -nayo, -neyo, -neureago, -neun, -neunda, -neundago, -neunde, -da, -daga, -dago, -daneun, -deogunyo, -deon, -dorok, -seumnida, -ja, -janeun, -jamaja, -ji, -jiman
-V	-n, -nde, -l, -m, -myeo, -myeon, -myeonse, -a, -ado, -aseo, -aya, -ayo, -at, -eo, -eodo, -eoseo, -eoya, -eoyo, -eot, -eotseot, -euna, -euni, -eureo, -eureogo, -eureoneun, -eumyeo, -eumyeon, -eumyeonse, -euseyo, -eusi, -eun, -eul, -eum

Suffix inventories in the Google Treebank Corpus (romanized)

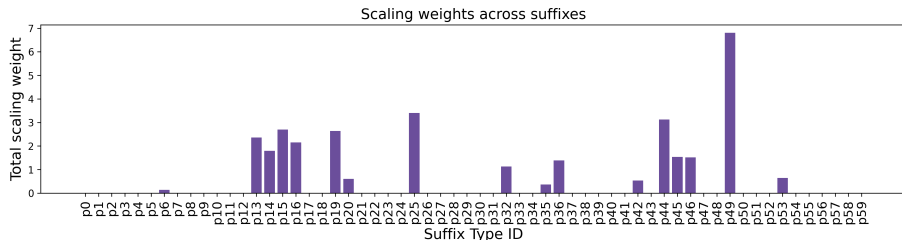
What seems to matter is **paradigm saturation** (Kodner, 2022)

- Naturalistic input is highly Zipfian: few stems occur across many suffixes, while most occur with only a few.
- Only a small subset of stems received nonzero scaling weights, and even fewer acquired substantial scaling.



What seems to matter is **paradigm saturation** (Kodner, 2022)

- In contrast, scaling factors were more widely assigned to suffixes, reflecting their broader attestation across stems.



To fully account for human language acquisition, two challenges remain:

- How can statistically non-dominant repair strategies be learned in the presence of overwhelmingly dominant patterns?
- How can the model correctly lexicalize the locus of exceptionality under conditions of limited paradigm saturation?

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Thank you for listening! 😊

Questions or comments?

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